The half-life t_h is the time corresponding to $A = 100\text{mg}$.

Thus, $t_h = \dfrac{2\log\left(\frac{1}{2}\right)}{\log\left(\frac{3}{4}\right)}$.

10.8.3. Newton's Law of cooling/warming



Consider pouring a $150°$ cup of coffee and kept it on the table in an $80°C$ room. What happens to the temperature of the coffee? We observe that the cup of coffee will cool off until it reaches the room temperature.



Now consider taking a $35°$ glass of cold water from the refrigerator and kept it on the table in an $80°C$ room. What happens to the temperature of the cold water? Similarly, we can observe the water will warm up until it reaches room temperature.

According to **Newton's law of cooling or warming**, the rate at which the temperature of a body changes is proportional to the difference between the temperature of the body and the temperature of the surrounding medium the so-called **ambient temperature**. If $T(t)$ represents the temperature of a body at time t, T_m the temperature of the surrounding medium, and $\dfrac{dT}{dt}$ the rate at which the temperature of the body changes, then Newton's law of cooling(or warming) is $\dfrac{dT}{dt} \propto T - T_m$ or $\dfrac{dT}{dt} = k\left(T - T_m\right)$, where k is constant of proportionality.

In either case, cooling or warming, if T_m is constant, it stands to reason that $k < 0$.

Example 10.29

In a murder investigation, a corpse was found by a detective at exactly 8 p.m. Being alert, the detective also measured the body temperature and found it to be 70°F. Two hours later, the detective measured the body temperature again and found it to be 60°F. If the room temperature is 50°F, and assuming that the body temperature of the person before death was 98.6°F, at what time did the murder occur?

$\left[\log(2.43) = 0.88789;\quad \log(0.5) = -0.69315\right]$

Solution

Let T be the temperature of the body at any time t and with time 0 taken to be 8 p.m.

By Newton's law of cooling, $\dfrac{dT}{dt} = k\left(T - 50\right)$ or $\dfrac{dT}{T - 50} = k\,dt$.

Integrating on both sides, we get $\log\left|50 - T\right| = kt + \log C$ or $50 - T = Ce^{kt}$.

When $t = 0,\ T = 70$, and so $C = -20$

When $t = 2, T = 60$, we have $-10 = -20e^{k2}$.

Thus, $k = \dfrac{1}{2}\log\left(\dfrac{1}{2}\right)$.

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